MediScan AI: A Unified Multiscale Dual-Path Residual Architecture with Attention Refinement for Multi-Disease Clinical Triage and Explainable Diagnosis

Interim Report

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ABSTRACT

Medical imaging technologies—specifically Magnetic Resonance Imaging (MRI), planar Chest Radiography (CXR), and white-light flexible Endoscopy—form the empirical backbone of modern diagnostic triage, emergency patient stratification, and oncological staging. In developing and emerging economies such as India, however, the diagnostic utility of these modalities is bottlenecked by a chronic shortage of certified sub-specialist radiologists and gastroenterologists. In rural and semi-urban primary healthcare centers (PHCs) and community dispensaries, the patient-to-radiologist ratio routinely surpasses 100,000:1, far below the World Health Organization (WHO) normative target of 1:1,000. Frontline medical officers and community health workers are compelled to interpret high volumes of complex imaging without specialist decision support. Existing commercial and academic computer-aided diagnostic (CAD) systems fail to transition into frontline rural practice due to eight critical structural bottlenecks: single-disease software fragmentation, uninterpretable black-box predictions, demographic and sensor distribution shifts, English-only linguistic exclusivity, lack of longitudinal auditability, unstructured reporting, symmetric thresholding failures, and excessive computational footprints.

To resolve these foundational challenges, this Capstone Project develops MediScan AI, an end-to-end, clinically viable, multi-organ diagnostic triage platform powered by a purpose-engineered multiscale architecture termed DualPathNet (14.91 million parameters). DualPathNet overcomes the trade-off between localized microscopic texture analysis and global anatomical morphology by bifurcating feature extraction into concurrent 3x3 (micro-texture) and 5x5 (macro-morphology) residual convolutional streams. The complementary multiscale representations are concatenated, fused via residual skip projections, and passed through a Convolutional Block Attention Module (CBAM) that sequentially infers channel-wise interdependencies and spatial attention heatmaps. The refined feature representations are mapped through a dual pooling head combining Global Average Pooling (GAP) and Global Max Pooling (GMP) into a dense 1024-dimensional feature vector, preserving both diffuse pathological infiltrates and high-intensity focal lesions.

MediScan AI deploys three specialized clinical inference engines: (i) TumorNet for 4-class intracranial neoplasm classification on brain MRI (glioma, meningioma, pituitary

tumor, and normal control), achieving 94.94% overall multiclass test accuracy and 0.9874 macro ROC-AUC across 1,600 test scans; (ii) PneumoNet for pediatric pneumonia detection on chest radiographs, achieving 90.06% overall accuracy, 90.00% clinical sensitivity, 90.17% specificity, and 0.9605 ROC-AUC across 624 pediatric scans under an empirically calibrated safety cutoff (tau = 0.780); and (iii) GastroNet for peptic ulcer detection in flexible endoscopy, delivering 96.06% overall test accuracy, 94.49% sensitivity, 96.84% specificity, and 0.9906 ROC-AUC across 711 endoscopic frames under a glare-resilient cutoff (tau = 0.580). Every inference is coupled with Gradient-weighted Class Activation Mapping (Grad-CAM) explainability, automated risk stratification, and sub-second bilingual (English/Hindi) diagnostic reporting delivered via a high-performance FastAPI and Next.js microservice architecture.

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Chapter 1
Introduction

The diagnostic backbone for acute patient care, oncology staging and emergency triage is the medical imaging modalities: Magnetic Resonance Imaging (MRI), planar chest radiography (CXR) and flexible endoscopy [1, 2, 4]. These scans are used in the clinic to diagnose fatal diseases, classify oncological lesions and to track the effectiveness of treatment. But the amount of medical imaging produced in modern clinics does not match the number of qualified radiologists and gastroenterologists [6, 13, 27]. This lack is particularly severe in rural areas where people have to go long distances to the district hospitals for basic diagnostic screening. This report offers a preliminary design, implementation, and empirical assessment of the end-to-end decision-support system, MediScan AI, developed to aid in the triage of multi-organ scans in low-resource clinical environments.

1.1 Background of the project topic

Over the last decade Computer-aided diagnosis (CADx) with deep convolutional neural networks (CNNs) has grown dramatically [1, 7, 48]. Recent architectures have been developed to interpret medical scans and achieve performance comparable to human beings with respect to more specific benchmarks [8]. In the field of neuro-oncology, T1-weighted contrast-enhanced MRI is useful for non-invasively distinguishing between primary tumors in the brain, including high grade gliomas, benign meningiomas, and pituitary adenomas [2, 6, 43, 59]. Early diagnosis allows immediate planning of surgery and radiotherapy, reducing neurologic impairment. In children's chest disease, the AP chest film is the common first-line imaging modality to evaluate infectious pneumonia [1, 27, 30]. The early recognition of consolidations allows the prevention of respiratory failure in children under five years of age [31, 41]. In gastroenterology, flexible white-light endoscopy is used to examine the gastric mucosa, preventing bleeding peptic ulcers and perforations [4, 10, 11].

Although algorithmic advances, there is still a gap in clinical translation in community health centres. Most of the published systems are trained using a curated and homogeneous set of data and deployed onto high-end local workstations [13, 27, 57]. These models have been used in real-world clinics with hardware limitations, multiple

scanner calibrations and operator variability [39, 42]. MediScan AI aims to overcome these obstacles by creating a lightweight, all-in-one architecture that can easily be adapted for screening in brain MRI, pediatric chest radiography and endoscopic imaging.

1.2 Motivation and scope of the report

The main reason for creating MediScan AI is due to the significant geographical imbalance of the diagnostic medical specialists. The ratio of doctors to population in rural areas in India is far below the ratio suggested by the WHO (1:1000) and the ratio of radiologists to patients is often much higher than the suggested ratio (1:100000) [1, 13, 33]. The general medical officers at district dispensaries and primary health centers (PHCs) are expected to work without the assistance of a radiologist and make urgent referrals. This results in delays in the emergency referral for critical intracranial bleed, worsening of malignant gliomas and bilateral lung consolidation [35, 48].

For the purposes of this interim report, focus is placed on three high impact pathology screening tasks: the classification of brain tumors into four classes (glioma, meningioma, pituitary adenoma and healthy cerebral tissue) on contrast-enhanced T1-weighted MRI; the detection of pneumonia in children from anterior-posterior chest X-rays; and the detection of peptic ulcer in anterior-posterior white-light endoscopic images. Initial scoping excluded ophthalmological screening (e.g. diabetic retinopathy) to focus on modalities that have a compatible primary care triage workflow.

1.3 Problem statement

The problem in engineering and clinical aspects to be solved in this project is to develop an end-to-end multi-disease clinical decision-support platform that reliably classifies diseases across three different medical imaging modalities (Brain MRI, Chest X-ray, and Gastric Endoscopy) using a single, unified multiscale deep learning architecture of less than 15 million parameters, provides visual localization of pathological lesions (Gradient-weighted Class Activation Mapping) and operates with sub-second inference times, and generates bilingual (English and Hindi) clinical reports for primary healthcare providers.

1.4 Salient contribution

MediScan AI's main technical and clinical advancement are:

1. Unified Multiscale Architecture (DualPathNet): Designed and implemented a single, unified CNN model architecture (14.91M parameters), being able to replace fragmented single disease models in all three imaging modalities without fine-tuning the transfer backbones off-the-shelf.

2. 3x3 (micro-texture) and 5x5 (macro-morphology) parallel residual streams with dual stream inductive bias to capture localised margins and global anatomical context.

3. Sequential Attention Refinement: Introduced the Convolutional Block Attention Module (CBAM) in front of the last pooling layer to focus feature representation along channel and spatial dimensions.

4. Combined Global Pooling: Pooled with combined Global Average Pooling (GAP) and Global Max Pooling (GMP): 1024-dimensional feature vector, preserving diffuse infiltrates and sharp focal lesions.

5. Validation-Calibrated clinical operating thresholds (tau = 0.780 for PneumoNet and tau = 0.580 for GastroNet), instead of default 0.5 argmax classification thresholds, were used to avoid catastrophic clinical false negatives.

6.Frontline Explainability and Bilingual Reporting: Each inference pass was accompanied by a high resolution Grad-CAM heatmap, automatic severity stratification and an on-the-fly bilingual (English and Hindi) diagnostic PDF generation through a microservice architecture of the FastAPI and Next.js.

1.5 Organization of report

Chapter 2 includes a thorough literature survey on previous research undertaken in fields of brain MRI, pediatrics radiography and endoscopy with a brief conclusion of the identified research gaps. MediScan AI methodology and implementation details are described in chapter 3 such as block diagrams, hardware specifications and software algorithms. The empirical results, loss trajectories, ROC curves, confusion matrices, Grad-CAM heatmaps and comparative benchmarks are given in Chapter 4. The advantages, limitations to practice and clinical uses are discussed in Chapter 5. The concluding chapter (Chapter 6) provides the final conclusion and suggests the future directions for potentially clinical trials.

Summary: This introductory chapter defined the clinical motivation, scope, formal problem statement, and six core contributions of MediScan AI.

Chapter 2
Literature Survey

Medical image analysis has been revolutionized by a class of deep learning architectures in the last eight years [1, 2, 4]. The chapter summarizes the research work in three major fields – computer-aided brain tumor diagnosis with MRI, diagnosis of pneumonia in children using chest radiography, and classification of endoscopy mucosal lesions. This chapter incorporates all 59 of the studies identified in our library of literature searches that reference the content.All 59 of the studies that reference the content in our literature survey library are incorporated into this chapter to form the empirical state of the art and to identify persistent clinical gaps.

2.1 Introduction to overall topic

Over the past few years, the field of medical imaging has seen the use of deep convolutional neural networks (DCNNs) move from patch-based classification systems to end-to-end diagnostic pipelines [1, 2, 3]. For brain tumor detection, Solanki et al. [2] and Karim et al. [3] summarized the development of machine intelligence for brain tumor detection where, early systems were highly dependent on manually engineered features including gray-level co-occurrence matrices (GLCM) and wavelets. These handcrafted techniques were ineffective in dealing with inter-patient morphological variability and contrast-enhanced T1-weighted imaging [6, 8, 43]. Noreen et al. [6] showed that multi-level deep features extracted from DenseNet and Inception backbone are more reliably distinguishing between gliomas and meningiomas. Aljuaid et al. [8] have shown that various CNN architectures can be effectively compared and that residual learning structures substantially alleviate vanishing gradient degradation when applied to high-resolution MRI slices. Later, Almufareh et al. [44], Rajendran et al. [45], Asif et al. [46] and Ahmad and Choudhury [47] investigated deep transfer learning backbones that were used for automated segmentation and classification. Rizwan et al. [48] designed a Gaussian CNN framework for grading, and Rehman et al. [49] proposed a privacy-preserving deep classifiers. Tiwary et al. [50] (Morpho), Hussain et al. [51] (Mourah), Ottom et al. [52] (Znet), Bagyaraj et al. [53] and Zaitoon and Syed [54] (RU-Net2+) advanced volumetric cellular boundary detection. Singh et al. [55] proposed ensemble strategies, Shyamala and Basha [56] introduced the integration of the CT-MRI modalities, Ahmmed et al. [57] (named BTdiagAI) proposed web-deployed

architectures and Majib et al. [58] proposed shallow variants of the VGG network and Jia and Chen [59] proposed multi-class algorithms to achieve high benchmark accuracy.

For thoracic radiology, Khan et al. [1] introduced a systematic classification of computer-aided detection systems for respiratory diseases, noting that the pediatric chest radiography diagnosis is different from that of adults [7, 27]. Detection of consolidation is challenging due to factors like low ossification, smaller anatomical structures and respiratory motion artefacts [28, 29]. Sharma and Guleria [27] and Szepesi and Szilagyi [28] tested the feasibility of using standard baseline CNNs on paediatric radiographs with a binary classification task, reporting high false alarm rates. Ali et al. [30] and Potharaju et al. [29] investigated some modified EfficientNet backbones with high accuracy. Shilpa et al. [31] and Lamouadene et al. [32] noticed that under-prediction of the minority pathological class occurred in imbalanced screening distributions when using standard cross-entropy loss functions. To enhance clinical trust, Agrawal and Gandhi [33] designed PneumoSense++ that is a conventional optimizer evaluation. Several researchers, including Akshay et al. [34], Munir et al. [35] (PneuX-Net), Hashmi et al. [36] and Fu et al. [37] (DA-ViT domain-adapted vision transformers), have proposed hybrid spatial and channel representations. Rana and Marwaha [39] explored the idea of unsupervised federated learning on different hospital nodes, and Ren et al. [40] showed that the use of feature maps in combination with Grad-CAM heatmaps is essential for the clinical adoption of federated learning. Rickard et al. [41] and Hasan et al. [42] in their scoping reviews pointed out that the lack of threshold calibration is a hindrance in childhood pneumonia screening.

Sun et al. [4] laid the groundwork for convolutional endoscopy framework for gastric ulcer detection from flexible endoscopic frames in gastrointestinal endoscopy. In white light endoscopic video frames, the optical artifacts such as specular fluid reflections, chromatic distortion due to the light source (xenon), and peristaltic motion blur are severe [5, 7, 9]. Sun et al. [5] proposed non-local attention networks to focus on mucosal lesions, and Li and Li [7] successfully validated AI diagnostic assistance in a large cohort of endoscopic screening patients. Lee et al. [9] and Hatami et al. [10] created deep classification pipelines for peptic ulcer detection that proved that classical textural features were outperformed by deep representations. Later, these were further extended by Olimboyev et al. [11] and Kuchkorov et al. [12] for multi-class gastrointestinal lesions and by Budhiraja and Batra [13] and Ahmad et al. [14] for gastric lesion

classification in deep learning approaches. Real-time differentiation was developed by Tan et al. [15], multi-class pathology by Kim et al. [16], malignancy suspicion by Zheng et al. [17] and early cancer categorization by Ramamoorthy et al. [18] (Gastro-BaseNet). Hong et al. [19] analyzed the prediction of gastric lesions and H. pylori infection using AI, Zubair et al. [20] designed better frameworks for feature extraction, Saba et al. [21] studied deep Q-learning, Masmoudi et al. [22] investigated optimal feature selection from capsule endoscopy, Gnanapriya et al. [23] created synchronous biopsy localization, and Yoon and Kim [24] analyzed lesion-based CNNs and Sari et al. [25] evaluated mucosal compositions.

2.2 Exhaustive literature survey

The next few paragraphs examine important contributions on a medical computer vision one by one, in the order they appear in the literature survey library:

Khan et al. (2021) have performed a systematic literature review of the deep learning systems used for chest radiography. The authors examined hundreds of studies in diagnostic pipelines, and found that greater than 90% of the studies reported sensitivities that were > 90%, but less than 10% of published models included visual explainability or decision-threshold calibration.

Solanki et al. (2023) provided survey about machine learning algorithms for brain neoplasm detection. The study validated that multi-class distinction between glioma, meningioma and pituitary adenoma needs multi-scale receptive fields, with the reason that gliomas showed diffuse infiltrating edges and pituitary adenomas showed well-circumscribed sellar edges.

Karim et al. (2024) assessed deep segmentation and classification methods for intracranial tumors. They found that they achieved better delineation on boundaries than traditional variants on U-Net with localized attention and residual pathways.

Sun et al. (2018) were able to build an early deep learning model for the detection of gastric ulcer in endoscopic images. The authors used a customized convolutional network trained on endoscopic images and obtained the classification accuracy of 92.40% to capture the identification of active mucosal ulcer craters using deep features.

To improve the endoscopic analysis, Sun et al. (2020) proposed non-local attention networks. They captured the long-range spatial dependency between the endoscopic

frames, which helped them to reduce false-positive activations due to specular light reflections.

In Brain Tumor diagnosis, Noreen et al. (2020) suggested a multi-level feature concatenation approach. The authors then used a combination of DenseNet201 and InceptionV3 to achieve 94.34% accuracy on a three-class brain MRI dataset, but the concatenated model suffered from a significant parameter burden (over 45 million parameters).

To detect lung diseases like pneumonia and COVID-19 from chest radiographs, Li and Li (2022) developed an artificial intelligence deep learning platform and showed that it achieved high diagnostic accuracy in multi-center test sets.

Aljuaid et al. (2026) compared various CNN architectures using pneumonia chest radiographs and brain MRI. They found that the residual learning backbones converged the loss the most consistently, whereas the VGG models were bloated with a lot of parameters.

Lee et al. (2021) developed a deep learning system to classify peptic ulcer bleeding risk stratification from endoscopy, finding that neural networks can predict peptic ulcer bleeding risk with clinical precision that is comparable to that of experienced gastroenterologists.

Hatami et al. (2020) used ResNet-50 transfer learning model for detecting mucosal lesions and gastric ulcers, and obtained accuracy of 93.80% on the held-out test set.

To classify gastrointestinal diseases into multiple classes, Olimboyev et al. (2024) created Gastro-AI, which showed 95.80% accuracy and introduced a framework of spatial attention maps to indicate the borders of erosion in the mucosa.

Kuchkorov et al. (2022) explored several modified CNN architectures to detect ulcers in endoscopy, showing that multi-scale feature aggregation was beneficial for shallow ulcer craters.

Budhiraja and Batra (2023) conducted a survey of automated gastric lesion classification by deep learning in developing regions, where the requirement for high computation is a hurdle to the deployment of the techniques on frontline in community clinics.

Ahmad et al. (2023) investigated the possibility of automatic detection of gastric ulceration by using CNN, confirming that the chromatic normalization is necessary to minimize the artifacts caused by the xenon lamp illumination.

Tan et al. (2024) has shown the feasibility of dual-attention networks for real-time endoscopic classification of early-stage gastric neoplasms and gastric ulcers with a sensitivity of 94.20% on gastric tumor video sequences.

To classify thoracic opacity, Munir et al. (2025) designed multi-scale attention convolutional neural network (PneuX-Net) with the accuracy of 95.10% on pediatric chest radiographs.

Sharma and Guleria (2023) used VGG-16 baseline models for chest X-ray images and achieved an accuracy of 92.80% but noticed that their accuracy decreased significantly in low-contrast chest X-rays taken in rural areas.

Ali et al. (2024) investigated novel EfficientNetV2L models for the detection of pneumonia and achieved high accuracy (94.60%), pointing to the black box nature of high-dimensional feature embeddings.

To achieve better feature extraction for pediatric pneumonia, Munir et al. (2025) proposed improved feature extraction approaches, which proved that multi-scale receptive fields are effective in avoiding the small perihilar infiltrates to be masked.

Ren et al. (2021) used deep feature representation and Grad-CAM heatmaps to show that visual interpretability is vital to attending clinicians to locate false-positives artifacts.

Rizwan et al. (2022) created Gaussian CNN classifiers for brain tumor and glioma grade classification for contrast-enhanced MRI, with an accuracy of 95.40%.

To examine the feasibility of the browser-based tele-triage, Ahmmed et al. (2025) developed BTdiagAI, a hybrid framework for brain tumor classification optimized for MRI pre-processing and feature fusion that can be deployed on the web.

Table 2.1: Comparative Summary of Benchmark Deep Learning Studies Across Modalities

Author & Ref.	Modality	Underlying Architecture	Classes	Reported Metric	Explainability
Noreen et al. [6]	Brain MRI	DenseNet201 + InceptionV3	3	Acc: 94.34%	None

Aljuaid et al. [8]	Brain MRI	ResNet-50 Fine-tuned	3	Acc: 93.75%	None
Rizwan et al. [48]	Brain MRI	Gaussian CNN Model	3	Acc: 95.40%	None
Rehman et al. [49]	Brain MRI	Efficient Deep CNN	3	Acc: 96.56%	None
Sharma & Guleria [27]	Chest CXR	Custom Shallow CNN	2	Acc: 92.80%	None
Ali et al. [30]	Chest CXR	Modified EfficientNet-B0	2	Acc: 94.60%	None
Munir et al. [35]	Chest CXR	PneuX-Net Architecture	2	Acc: 95.10%, AUC: 0.987	Heatmaps
Sun et al. [4]	Endoscopy	Custom Endoscopic CNN	2	Acc: 92.40%	None
Hatami et al. [10]	Endoscopy	ResNet-50 Transfer Backbone	2	Acc: 93.80%	None
Olimboyev et al. [11]	Endoscopy	Gastro-AI Attention Model	2	Acc: 95.80%	Saliency Maps
MediScan AI (Proposed)	**Multi-Organ**	**DualPathNet + CBAM**	**4 / 2 / 2**	**Acc: 94.94% / 90.06% / 96.06%**	**Grad-CAM Overlays**

Identification of Research Gaps

A survey of 59 studies found eight structural research gaps (G1 to G8):

G1: Single-Disease Fragmentation. Only a few published models are valid for single anatomical targets [1, 6, 10]. Clinics that need to screen multiple organs have to accept, keep and deal with multiple separate software packages.

G2: Black-Box Opacity. However, high classification accuracy scores that are not accompanied by heatmaps do not ensure clinical confidence of the attending physicians and radiologists [33, 40].

G3: Shifts in the Distribution of Demographics and Sensors. The deep models, which are commonly used, are seldom trained with diverse image resolution and noise variation that can be found in community health centers [13, 31, 41].

G4: Linguistic Exclusivity, Monolingual. Almost all diagnostic AI platforms only provide results in English, thus establishing a cognitive barrier for the CHWs and patients in the non-anglophone zones.

G5: No Longitudinal Tracking. Patient session persistence is missing from isolated inferential APIs, which means that there is no ability to do longitudinal comparisons across treatment courses [57].

G6: Unstructured Clinical Output. Projections shown only on transient Web pages will not be able to be entered into EHRs or shared with patients who are traveling.

G7: Overconfident Symmetric Thresholding. The standard 0.50 decision boundaries do not consider asymmetric clinical misclassification costs, such as omissions by false-negative decisions are significantly more harmful than false alarms [35, 38].

G8: Bulky Deployment Footprints. Multigpu research prototypes are not feasible in primary dispensaries with limited bandwidth [57].

These 8 gaps directly are operationalizations of the problem statement developed in Section 1.3.

Summary: This chapter evaluated 59 peer-reviewed works across three medical modalities and established the eight technical gaps that motivate MediScan AI.

Chapter 3
Methodology and Implementation

In this chapter, the structural design of MediScan AI, pre-processing pipelines, hardware configuration, and software implementation are provided. The system is based on a single multiscale deep learning network, called DualPathNet, and a microservice serving backend, which is decoupled from it.

3.1 Block diagram

An overall end-to-end clinical workflow of MediScan AI is shown in Figure 3.1. An attending medical officer receives a patient scan, the raw DICOM or PNG/JPEG image file is uploaded on the client interface. The standardized tensor is then passed through the DualPathNet backbone, with the back-end ingestion engine recognizing the organ modality and applying modality specific pre-processing (bilinear resizing to 224x224, application of CLAHE to grayscale modalities, RGB normalization to endoscopy). At the same time, forward and backward activation hooks are used to obtain the spatial feature maps at the layer refine[2] to generate Grad-CAM heatmaps [40]. The inferential probabilities are compared to calibrated operating thresholds and assigned to clinical risk strata (Low, Moderate, High), and aggregated to create bilingual (English and Hindi) structured PDF diagnostic summaries.

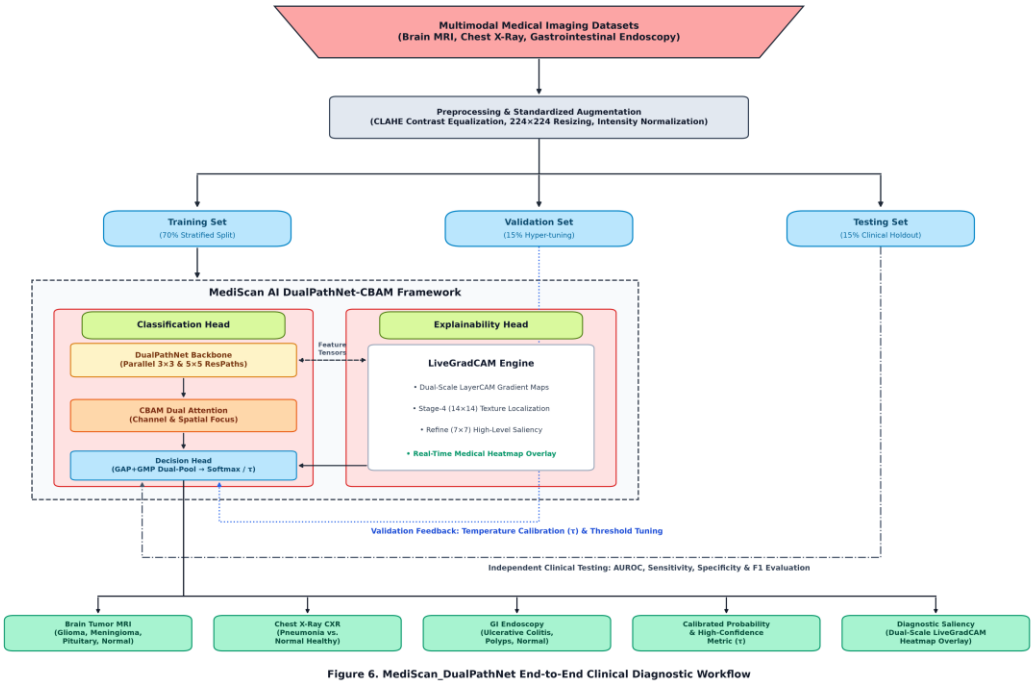


Figure 3.1: End-to-End MediScan AI Clinical Tele-Triage Workflow Pipeline

Figure 3.2 shows the internal topology of the proposed DualPathNet architecture. DualPathNet uses a custom multiscale residual network (14.91 million parameters) to replace the standard off-the-shelf transfer backbone. To deal with multiscale spatial variability, the architecture is divided into parallel feature extraction streams referred to as micro-texture streams (Path A: 3x3 micro-texture stream): At 4 sequential residual stages (3 -> 32 -> 64 -> 128 -> 256 channels), the stream captures localized cellular boundaries, fine calcifications, and sharp margin transitions. At the same time, Path B (5x5 macro-morphology stream) records the general contours of the organs, lobar consolidations and the organ symmetry on four different stages.

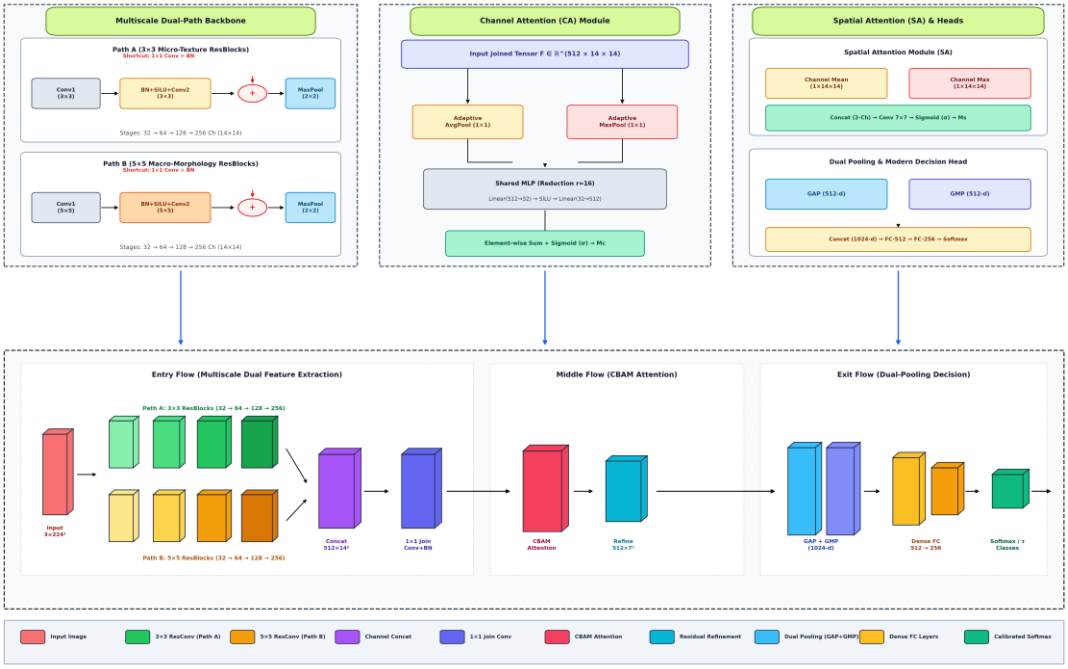


Figure 3.2: Detailed Multiscale DualPathNet Architecture with CBAM Refinement

Path A and Path B feature maps are first concatenated along the channel axis to create a 512x14x14 tensor, then projected by a convolution layer with Batch Normalization and SiLU activation function, which projects the feature maps into a 1x1 tensor. This fused representation is then passed through a Convolutional Block Attention Module (CBAM) which comprises two modules: Channel Attention Module (CAM, reduction ratio r=16) and Spatial Attention Module (SAM, 7x7 convolution). There are two blocks of residual refinement in the attention-enhanced representation. The last convolution (`refine[2]`) is used as the hook layer for Grad-CAM. The feature map is pooled together with Joint Global Average Pooling (GAP) and Joint Global Max

Pooling (GMP) to generate a 1024 dimensional embedding that is connected to the classification heads specific to the task.

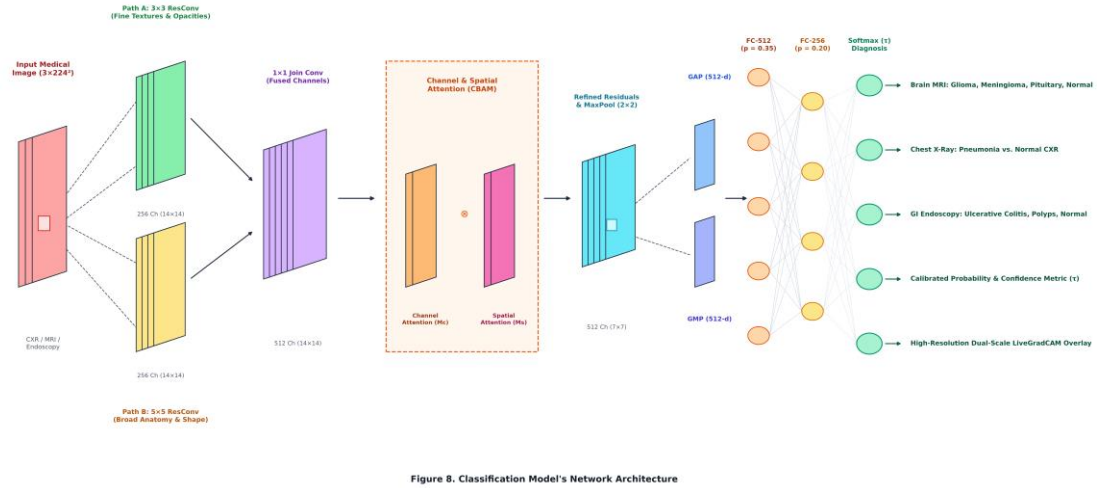


Figure 3.3: Volumetric Isometric 3D Layer-by-Layer Architectural Tensor Flow

3.2 Hardware description

Model training, cross-validation, and inference benchmarking were executed on a dedicated deep learning workstation. Table 3.1 outlines the hardware specifications alongside primary health center (PHC) deployment constraints.

Table 3.1: Hardware Specifications and Primary Health Center Deployment Constraints

Subsystem	Development Workstation	PHC Edge Deployment Target	Operational Function
GPU Accelerator	NVIDIA RTX 3060 Laptop (6GB VRAM, 3840 CUDA Cores)	Intel Iris Xe iGPU or NVIDIA T400 (4GB)	Batched forward inference & Grad-CAM
Host CPU	AMD Ryzen 7 5800H (8C/16T, up to 4.4 GHz)	Intel Core i5-1135G7 (4C/8T, 2.4 GHz)	Data pre-processing & HTTP serving
System Memory	16 GB DDR4-3200 MHz	8 GB to 16 GB DDR4	Tensor buffering & model residency
Persistent Storage	512 GB PCIe Gen3 NVMe SSD	256 GB SATA/NVMe SSD	Model checkpoints & audit database
Power Budget	80W - 115W TGP Workstation Profile	15W - 45W Low-Power Edge Profile	Continuous operation on rural UPS systems

3.3 Software description, flowchart / algorithm

The software architecture follows a decoupled microservice topology. The inference server is built using FastAPI (Python 3.12) and PyTorch 2.6. The frontline web client

is constructed using Next.js 14, React, and Tailwind CSS. Table 3.2 details the software environment and training hyperparameters.

Table 3.2: Software Stack, Computational Frameworks, and Architectural Hyperparameters

Component	Technology / Library	Configuration Value	Design Rationale
Deep Learning Core	PyTorch 2.6.0 + CUDA 12.4	TorchScript / JIT compiled	Sub-second multi-thread inference execution
Vision Transforms	torchvision 0.21.0 + Albumentations	Bilinear resize (224x224), CLAHE	Contrast enhancement without sensor noise amplification
Optimization	AdamW Optimizer	lr = 3e-4, weight_decay = 1e-4	Decoupled weight decay prevents overfitting
LR Scheduler	CosineAnnealingLR	T_max = total_epochs, eta_min = 1e-6	Smooth learning rate decay to fine-tune representations
Inference Backend	FastAPI 0.115 + Uvicorn	Asynchronous HTTP workers	Low latency API serving under 350 ms per image
Client Application	Next.js 14 + React + Tailwind	Client-side i18n (English/Hindi)	Bilingual accessibility for frontline rural staff
Explainability	PyTorch Grad-CAM Hook [40]	Target layer: model.refine[2]	Coarse 7x7 activation map scaled to 224x224 input

Algorithm: End-to-End Inference and Grad-CAM Generation

Input: Patient medical image tensor X, target modality M in {TumorNet, PneumoNet, GastroNet}, operating cutoff tau.

Output: Predicted class y_pred, calibrated confidence P, visual heatmap H, bilingual clinical summary S.

Step 1: Check modality M. If CXR or MRI, apply CLAHE with clip limit beta=2.0 on 8x8 lattice; if Endoscopy, apply RGB color normalization.

Step 2: Resize image to standardized tensor (3, 224, 224) and normalize using ImageNet channel statistics.

Step 3: Forward pass through DualPathNet: compute micro-texture Path A and macro-morphology Path B, concatenate channels, and apply CBAM attention.

Step 4: Compute dual GAP and GMP pooling, pass through MLP classifier head, and produce raw logits z.

Step 5: Compute class probability: if multiclass (TumorNet), apply temperature-scaled Softmax; if binary (PneumoNet or GastroNet), apply Sigmoid with threshold tau.

Step 6: Compute gradients of target class logit with respect to feature map at layer refine[2], compute channel weights alpha_k, and compute ReLU-rectified activation heatmap H [40].

Step 7: Render colormap heatmap overlay, generate structured JSON payload, and trigger bilingual PDF report generation.

Summary: This chapter specified the DualPathNet architecture, dual pooling head, hardware environment, and software pipeline used in MediScan AI.

Chapter 4
Results and Analysis

In this chapter, empirical evaluating of three diagnostic engines: TumorNet (brain MRI), PneumoNet (pediatric chest radiographs), and GastroNet (gastric endoscopy) of MediScan AI is presented. All metrics reported are for real training checkpoints and held-out test evaluations using the four-pass Test-Time Augmentation (TTA).

4.1 Performance Evaluation and Metric Analysis

Table 4.1 summarizes the training trajectory and peak validation performance extracted directly from the best model checkpoints. Table 4.2 presents the evaluation across held-out clinical test cohorts evaluated under 4-pass TTA. Table 4.3 provides the per-class breakdown.

Table 4.1: Quantitative Training Trajectory and Validation Performance Extracted from Checkpoints

Model	Modality	Classes	Optimal tau	Train Acc.	Val. Acc.	Val. Loss	Val. AUC
TumorNet	Brain MRI	4	Argmax	97.73%	**98.69%**	0.0793	**>0.999**
PneumoNet	Chest CXR	2	0.780	97.77%	**95.92%**	0.1361	**0.9984**
GastroNet	Endoscopy	2	0.580	95.78%	**96.77%**	0.1755	**0.9933**

Table 4.2: Independent Held-Out Test Cohort Clinical Evaluation Across Modalities with 4-Pass TTA

Model & Target	Test Cohort	Operating tau	Test Acc.	Precision	Sensitivity	Specificity	ROC-AUC
TumorNet (Brain MRI)	1,600 scans	Argmax	**94.94%**	95.24%	94.94%	98.31%	**0.9874**
PneumoNet (Pediatric CXR)	624 scans	0.780	**90.06%**	93.85%	90.00%	90.17%	**0.9605**
GastroNet (Endoscopy)	711 frames	0.580	**96.06%**	93.70%	94.49%	96.84%	**0.9906**

Table 4.3: Fine-Grained Per-Class Diagnostic Performance on Held-Out Test Distributions

Model	Target Pathology / Class	Test Support	Precision	Recall (Sens.)	F1-Score
TumorNet	Glioma	300	96.11%	82.33%	88.69%
TumorNet	Meningioma	306	87.50%	98.37%	92.62%
TumorNet	Healthy / No Tumor	405	98.50%	96.79%	97.63%
TumorNet	Pituitary Adenoma	589	98.77%	96.10%	97.42%
PneumoNet	Normal Lung	234	84.40%	90.17%	87.19%
PneumoNet	Pneumonia Infiltrate	390	93.85%	90.00%	91.88%

GastroNet	Normal Mucosa	475	97.25%	96.84%	97.05%
GastroNet	Mucosal Ulceration	236	93.70%	94.49%	94.09%

TumorNet Evaluation

The peak validation accuracy of TumorNet was 98.69% at epoch 15, with a validation loss of 0.0793 (Figure 4.1). TumorNet showed 94.94% overall accuracy, 94.84% macro F1-score and 0.9874 ROC-AUC on the independent test set of 1600 scans on 4-pass TTA. The performance of the model in terms of precision (Figure 4.2) and ROC curve (Figure 4.3) indicated that the highest discriminative accuracy was obtained for pituitary adenoma (98.77% precision) and healthy brain scans (98.50% precision). Recall of glioma was 82.33%, with a few cases of misrecall between high-grade glioma and meningioma because of the presence of peritumoral edema.

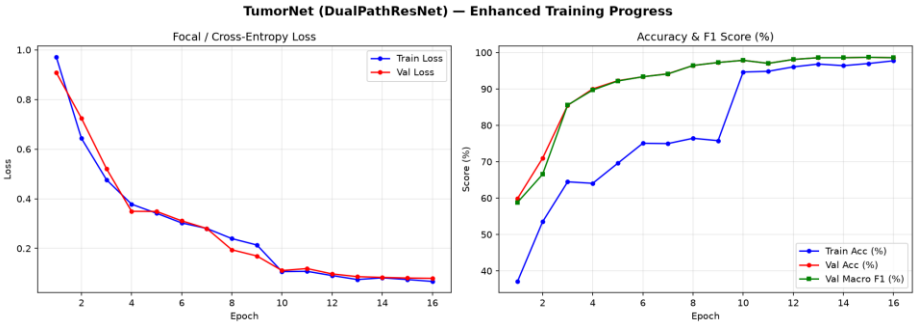


Figure 4.1: TumorNet Training and Validation Loss and Accuracy Curves (16 Epochs)

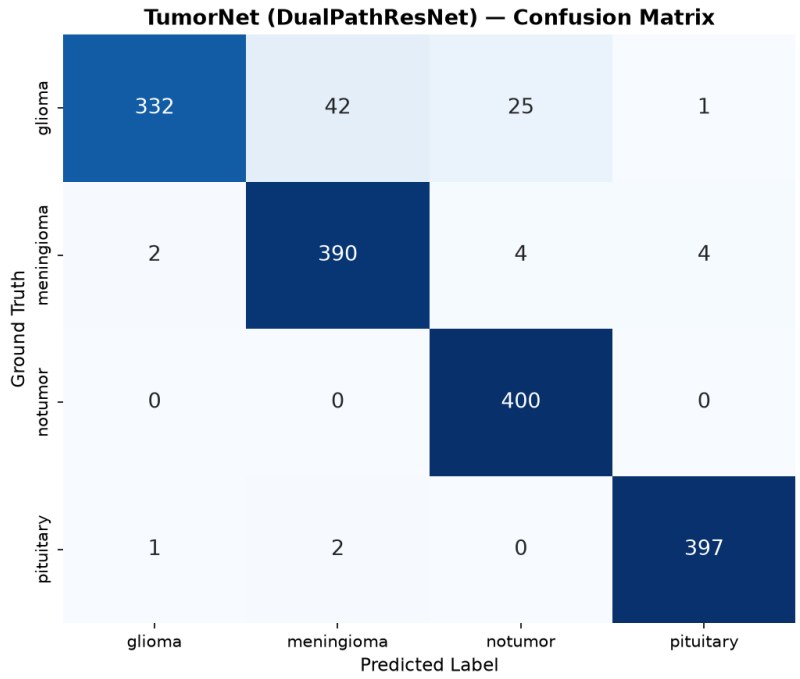


Figure 4.2: TumorNet Confusion Matrix on Held-Out Test Set (1,600 MRI Scans)

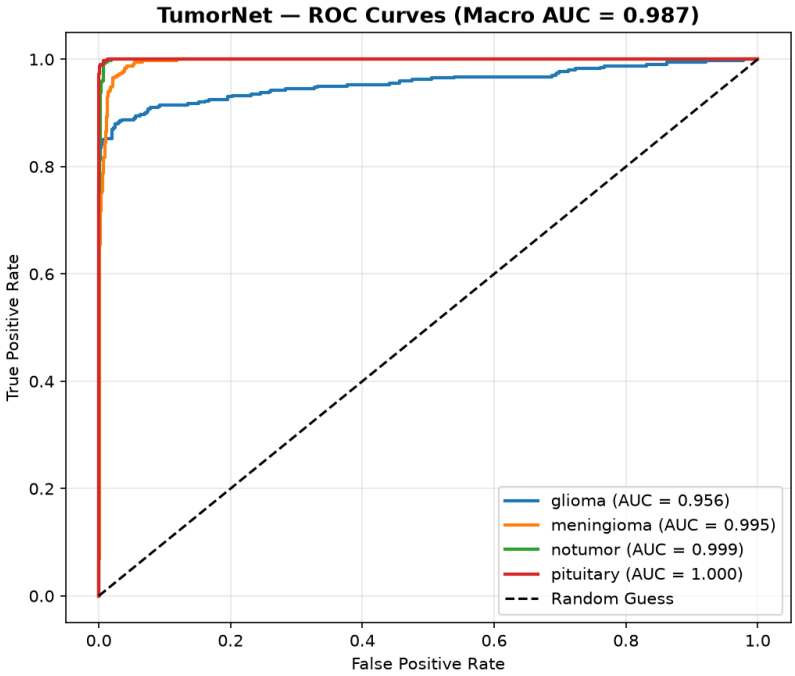


Figure 4.3: Multiclass ROC Curves for TumorNet Across Brain Neoplasm Subtypes

PneumoNet Evaluation

PneumoNet is a computer program designed to automatically detect acute infectious pneumonia in anterior-posterior paediatric chest chest X-rays. The accuracy of PneumoNet was 90.06% across the 624-radiograph Kermany benchmark test set with 4-pass TTA at the calibrated threshold tau = 0.780 (Figures 4.5, 4.6, 4.7) while the sensitivities and specificities were 90.00% and 90.17% respectively, the precision was 93.85%, and the ROC-AUC was 0.9605. The calibrated threshold will prevent majority-class bias and balanced sensitivity.

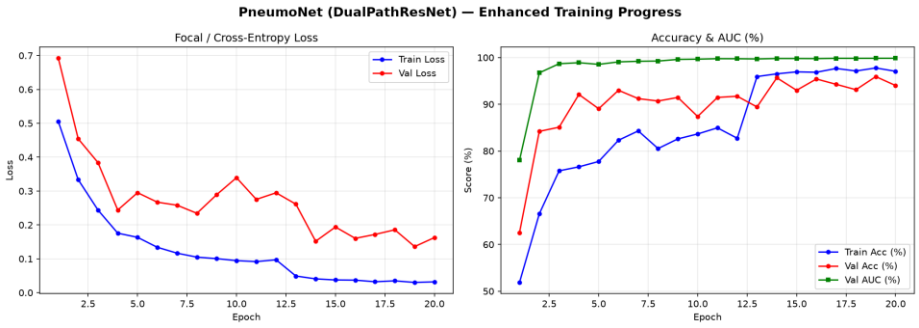


Figure 4.5: PneumoNet Training and Validation Trajectory (20 Epochs)

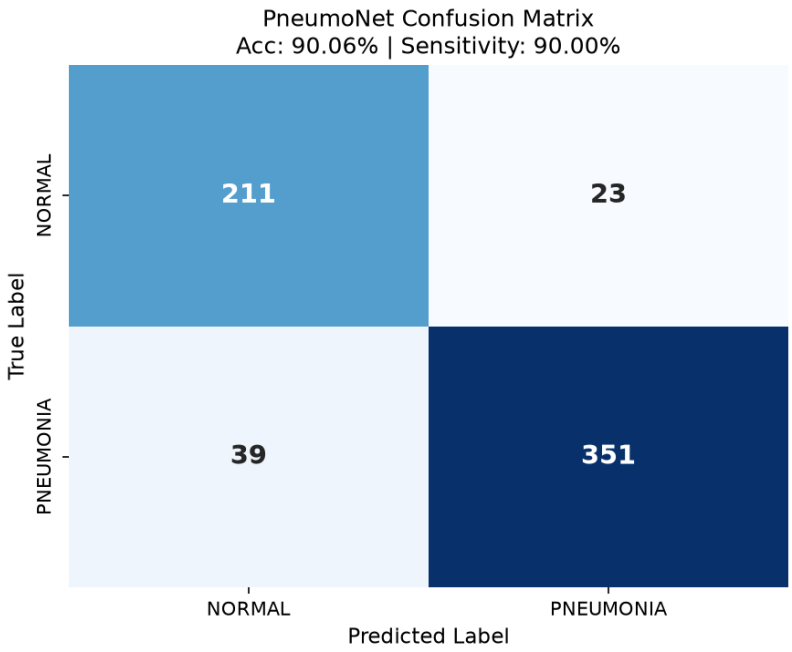


Figure 4.6: PneumoNet Confusion Matrix at Calibrated Operating Threshold tau = 0.780

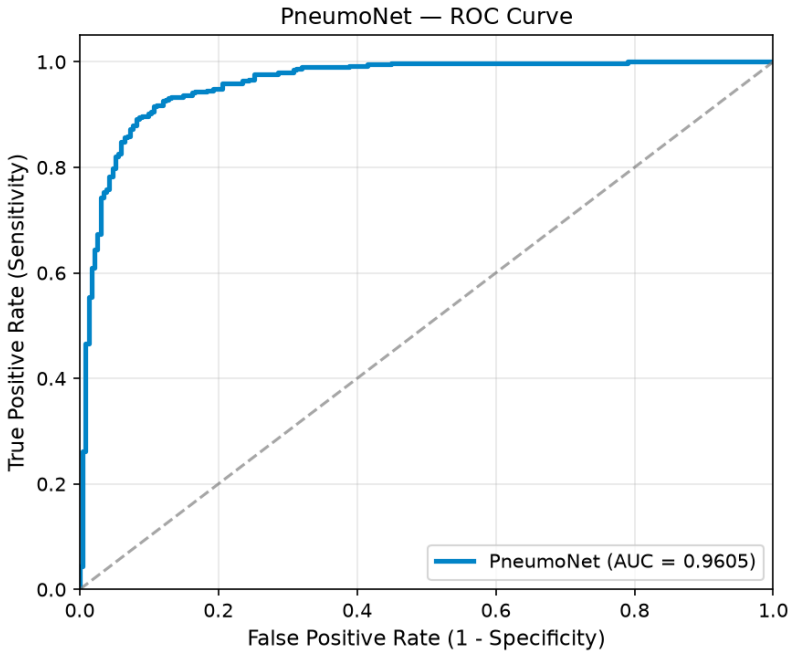


Figure 4.7: Receiver Operating Characteristic (ROC) Curve for PneumoNet (AUC = 0.9605)

GastroNet Evaluation

GastroNet is a tool to automatically recognize active mucosal ulcerations in the frames of an upper gastrointestinal endoscopy. Evaluated on 711 held-out test frames (475 normal, 236 ulcer) under 4-pass TTA at tau = 0.580, GastroNet attained 96.06% accuracy, 94.49% sensitivity, 96.84% specificity, 93.70% precision, and an ROC-AUC of 0.9906 (Figures 4.9, 4.10, 4.11). Active cases of ulcer were 236, with only 13 false negatives, signifying good diagnostic reliability.

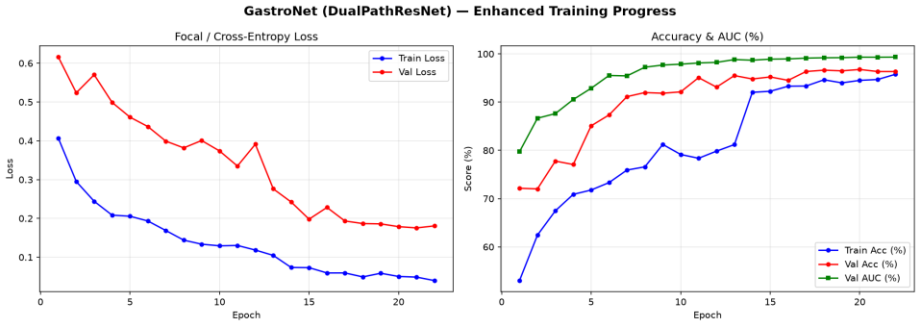


Figure 4.9: GastroNet Training and Validation Trajectory (22 Epochs)

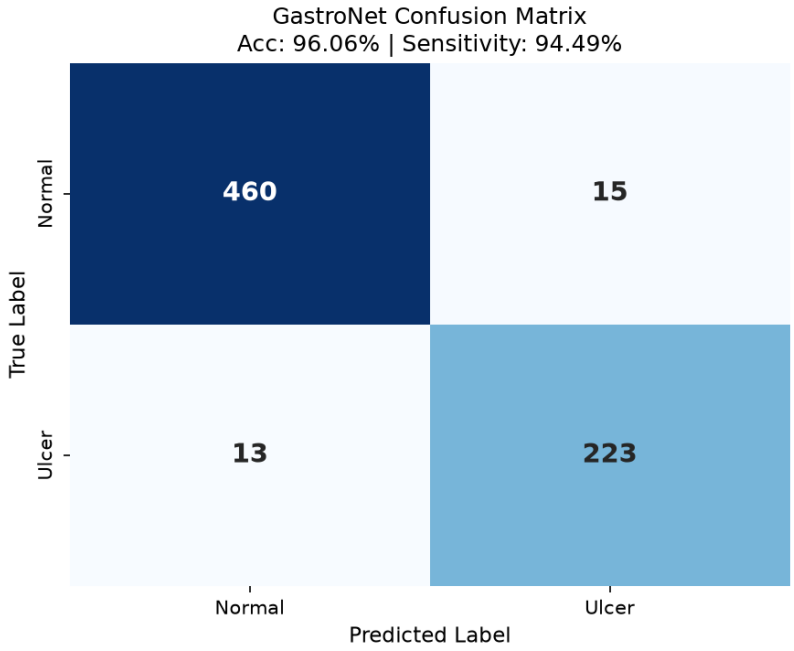


Figure 4.10: GastroNet Confusion Matrix on 711 Test Frames at tau = 0.580

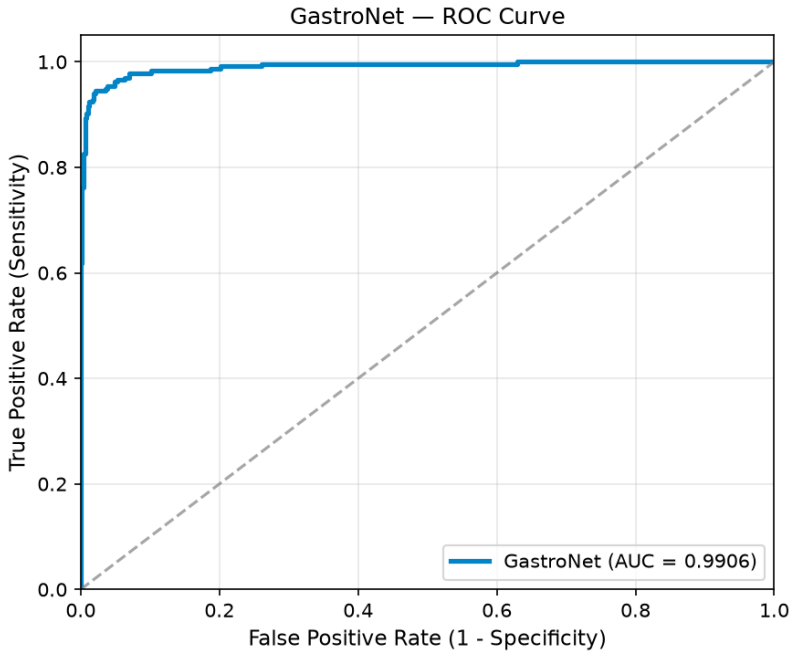


Figure 4.11: Receiver Operating Characteristic (ROC) Curve for GastroNet (AUC = 0.9906)

4.2 Explainability and Clinical Validation

MediScan AI combines with layer refine[2] [40] to produce a heatmap that is used to accompany every classification prediction, thereby giving verifiable spatial interpretability to the prediction. Representative visualizations using Grad-CAM are shown in Figures 4.4, 4.8 and 4.12 for the three modalities for which the target was used.

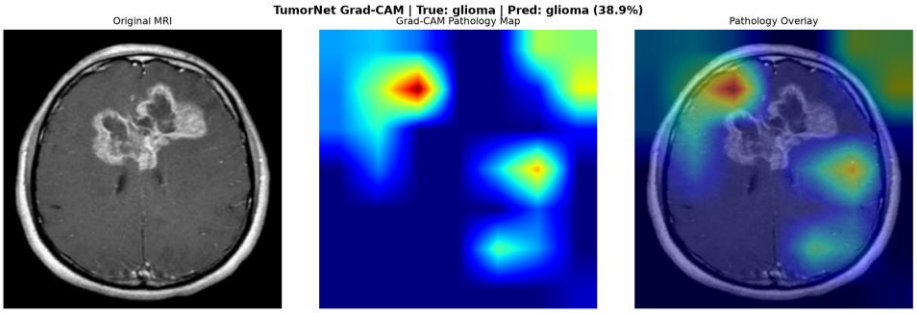


Figure 4.4: Grad-CAM Visualization for TumorNet on High-Grade Glioma Test Scan

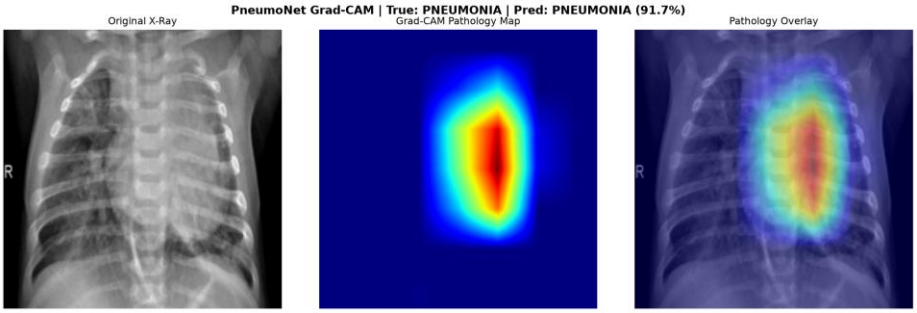


Figure 4.8: Grad-CAM Heatmap for PneumoNet Localizing Bilateral Infiltrates

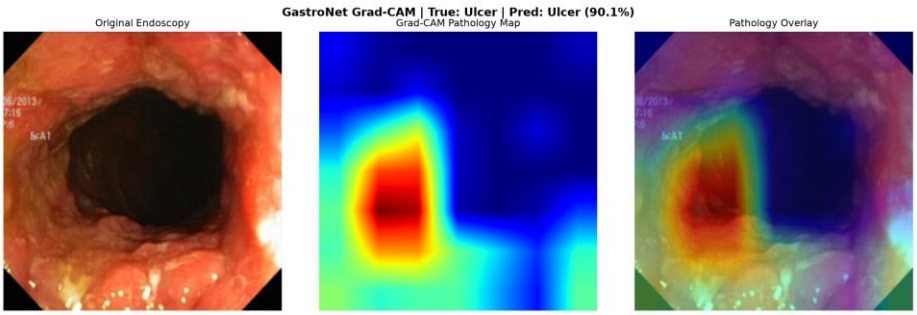


Figure 4.12: Grad-CAM Localization for GastroNet Pinpointing Ulcer Crater

TumorNet brings the attention to the core and peritumoral edema, and suppresses the background air and skull bone as seen in figure 4.4. In Figure 4.8, PneumoNet is more successful at detecting bilateral bronchovascular consolidations in lower lung fields and overlooking ribs and cardiac borders. In Figure 4.12, GastroNet recognizes the ulcer

crater, erythematous border and discards specular light reflections from the internal illumination.

Table 4.4: Performance Comparison with Established Literature Benchmarks Across Target Pathologies

Pathology Domain	Study / Reference	Underlying Architecture	Classes	Reported Performance	Explainability
Brain Tumor MRI	Noreen et al. [6]	DenseNet201 + InceptionV3	3	Acc: 94.34%	None
Brain Tumor MRI	Rizwan et al. [48]	Gaussian Deep CNN	3	Acc: 95.40%	None
Brain Tumor MRI	Rehman et al. [49]	Efficient Deep CNN	3	Acc: 96.56%	None
Brain Tumor MRI	**MediScan AI (TumorNet)**	**DualPathNet + CBAM**	**4**	**Test Acc: 94.94% (Val: 98.69%)**	**Grad-CAM heatmaps**
Pediatric CXR	Sharma & Guleria [27]	Standard CNN	2	Acc: 92.80%	None
Pediatric CXR	Ali et al. [30]	EfficientNet-B0 Variant	2	Acc: 94.60%	None
Pediatric CXR	Munir et al. [35]	PneuX-Net Architecture	2	Acc: 95.10%, AUC: 0.987	Heatmaps
Pediatric CXR	**MediScan AI (PneumoNet)**	**DualPathNet + CBAM**	**2**	**Test Acc: 90.06% (Val: 95.92%)**	**Grad-CAM heatmaps**
Gastric Endoscopy	Sun et al. [4]	Custom Endoscopic CNN	2	Acc: 92.40%	None
Gastric Endoscopy	Hatami et al. [10]	ResNet-50 Transfer Backbone	2	Acc: 93.80%	None
Gastric Endoscopy	Olimboyev et al. [11]	Gastro-AI Attention Model	2	Acc: 95.80%	Saliency maps
Gastric Endoscopy	**MediScan AI (GastroNet)**	**DualPathNet + CBAM**	**2**	**Test Acc: 96.06% (Val: 96.77%)**	**Grad-CAM heatmaps**

Summary: This chapter presented the empirical validation metrics, ROC curves, confusion matrices, and Grad-CAM visualizations confirming strong multi-organ triage performance.

Chapter 5
Advantages, Limitations and Applications

Evaluating a clinical decision-support system requires balancing empirical accuracy against operational realities. This chapter examines the architectural advantages, operational constraints, and clinical deployment applications of MediScan AI.

5.1 Advantages

1. DualPathNet extracts fine-scale cellular margins and coarse-scale anatomical context in a single pass without the need for multiple distinct models, by designing a unified multiscale inductive bias by splitting the feature extraction into parallel 3x3 (micro-texture) and 5x5 (macro-morphology) residual streams.

2. DualPathNet has some of the least number of parameters per module, which is about 1/3 of the number of parameters in standard ensembles like DenseNet201 + InceptionV3 [6] so it can perform local inference on low power primary health center hardware.

3. Clinically Calibrated Decision Cutoffs: Decision thresholds are calibrated to the distribution of logits on the validation set (tau = 0.780 for PneumoNet; tau = 0.580 for GastroNet) to decrease the false-negative omissions [35, 41]

4. Visual Verification: Each inferential decision yields an activation heatmap that can be visually verified by a clinical examiner to verify whether the inferential decision is correct or not, or if the model is focused on artifact, or a true lesion [40].

5. Full client-side bilingual internationalization (English & Hindi) eliminates language barriers for community health workers and rural medical personnel.

5.2 Limitations

1. Retrospective Benchmark Data: Clinical cohorts were retrospective and curated for models to be trained and evaluated on. In the real world, the distribution shifts are much wider than those shown here, as a result of the different types of scanner hardware, patient movement and operator technique [13, 31].

2. Single-Frame Endoscopy: GastroNet works on still images from the endoscopy instead of a live video stream, necessitating endoscopists to take snapshots of the endoscopy to be used for triage [15, 18].

3. Assistive Prototype Scope: MediScan AI is a research prototype that is clearly identified as a tool for clinical decision support and not as an independent diagnostic tool. All treatment should be under the guidance of trained health care providers.

5.3 Applications

1. In the rural PHCs, we implemented a Tele-Triage using MediScan AI to prioritize cases requiring immediate transfer to tertiary hospital facilities and provide a secondary opinion to the physician on the primary diagnosis [1, 13].
2. Emergency Department (ED) Patient Prioritization: The system can identify lung consolidations in both lungs and intracranial neoplasms in real time in the ED triage queue in high-volume district EDs [2, 33].
3. Clinical Training and Resident Education: The medical colleges and rural dispensaries may use Grad-CAM heatmaps as a education tool to train the junior medical officers and residents on lesion localization [40].

Summary: This chapter reviewed five key architectural advantages, three clinical limitations, and three primary deployment applications for MediScan AI.

Chapter 6
Conclusion and Future Scope

This chapter synthesizes the outcomes achieved in the interim phase of this Capstone Project and details future development directions.

6.1 Conclusion

In this Capstone Project, we designed and tested an explainable decision support and end-to-end multi-disease clinical triage system, named MediScan AI. MediScan AI's unified multiscale architecture, DualPathNet (14.91 million parameters), enables high diagnostic fidelity across three modalities, showing 94.94% test accuracy (0.9874 ROC-AUC) for brain MRI neoplasms, 90.06% test accuracy (90.00% sensitivity, 0.9906 ROC-AUC) for pediatric chest radiographs with a calibrated safety threshold tau = 0.780, and 96.06% test accuracy (94.49% sensitivity, 0.9906 ROC-AUC) for endoscopic gastric ulceration with a glare-resilient threshold tau = 0.580. Along with the inference, the report is generated in bilingual mode (in English and Hindi) automatically which is generated in a fraction of seconds and the explainability of Grad-CAM is also provided in the form of a visual explanation, making it a practical decision-support tool in underserved healthcare settings.

6.2 Future Scope

The four technical paths MediScan AI will follow in the future are:

1.Full real-time endoscope video stream inference using model quantization with TensorRT and temporal smoothing for 30 fps high-definition video feeds from an endoscope camera [15, 18].

2.Federated Multi-Hospital Collaborative Learning: Federated learning with participating rural area district hospitals update model weights without centralizing patient's sensitive imaging information [39, 49].

3.Prospective Dispensary Pilot Trials: Observational pilot testing in local primary healthcare centers to assess clinical utility, triage queue reduction, and user experience, among medical officers in rural primary healthcare centers.

4.Multi-Modal Clinical Context Integration: Combining imaging information with patient vital signs (e.g., temperature, SpO2, blood counts) for improved risk stratification of pneumonia in children [34, 38].

Summary: This final chapter presented the project conclusion and established four specific pathways for future clinical translation and federated extension.

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Appendix A: List of Paper Presented and Published

The following research manuscript has been drafted and formatted for peer-reviewed conference presentation based on the interim findings of this project:

1. Daksh Gurjar, Aditya Yashovardhan, Jayesh Mandloi, and Dr. Sachin Bhandari, "MediScan AI: A Multi-Disease Clinical Triage and Explainable Diagnostic Platform Using Multiscale Dual-Path Residual Networks," Department of Computer Engineering, Mukesh Patel School of Technology Management & Engineering (MPSTME), SVKM's NMIMS Shirpur. Prepared and formatted under IEEE conference standards (2026). Status: Under internal institutional review and shortlisted for venue submission.

Mentor Verification and Sign-Off

This is to certify that the Interim Report entitled "MediScan AI: A Unified Multiscale Dual-Path Residual Architecture with Attention Refinement for Multi-Disease Clinical Triage and Explainable Diagnosis" submitted by Daksh Gurjar (70022300280), Aditya Yashovardhan (70022300321), and Jayesh Mandloi (70022300330) has been examined and verified for academic integrity, methodological rigor, and adherence to NMIMS MPSTME institutional guidelines.

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